Cosmology with Fast Radio Bursts

LABORATOIRE IV THEORIQUE

Spring 2026

Instructor: Ajith Sampath

Contact : ajith.sampath@unige.ch

Project Introduction

This short semester project combines theoretical foundations, computational modeling, and hands-on analysis of real observational data. It is designed to guide students through the full workflow of modern astrophysical research—from understanding the underlying physical principles, to implementing cosmological models, to interpreting and validating results using actual measurements from contemporary surveys.

Fast Radio Bursts (FRBs) are extremely short, millisecond-duration radio pulses that originate from distant, often extragalactic sources. Instruments such as the CHIME radio telescope in Canada have detected hundreds of these events. As FRB signals travel through ionized gas, they experience dispersion: lower-frequency waves arrive later than higher-frequency ones. This dispersion carries information about the total number of free electrons encountered along the line of sight, and therefore about the baryonic matter distributed throughout the Universe.

The key observable is the dispersion measure (DM), which quantifies the integrated free-electron density between the source and the observer. In a flat ΛCDM cosmology, derived from the Friedmann equation, the expansion history is given by

$$H(z) = H_0\sqrt{\Omega_m(1+z)^3 + \Omega_\Lambda}, \tag{1}$$

which leads to a theoretical relationship between dispersion measure and redshift that depends mainly on the cosmic baryon density Ω_b. Although Ω_b is tightly constrained at recombination by the Planck mission, it is important to verify whether late-time measurements of baryonic matter remain consistent with these early-Universe constraints.

This project examines whether the observed DM–z relation of localized FRBs agrees with the predictions of ΛCDM and the constraints from Planck, and explores whether FRBs can act as an independent probe of the cosmic baryon distribution and large-scale structure.

I Background Reading

Understand the theoretical foundation of the topic by reading the following selected resources and there are much more resources online to read further.

- *Modern Cosmology* by Scott Dodelson.
- The famous FRB detection paper by Duncan and the recent review paper of FRBs.
- Slides [1] and [2].
- Papers [1], [2], [3] and [4] on FRB science (each of them covers basics and some advanced discussions).

II Theoretical framework and modeling

1. Derive Friedmann Equations and get the flat Λ-CDM form (as in Equation [1]) and explain the assumptions and conditions involved.
2. Derive the expressions for baryon density parameter (Ω_b) and its relation to the electron density (n_e) in the inter-galactic medium (IGM).
3. Derive the Macquart relation :
$$\mathrm{DM}_{\mathrm{IGM}}(z) = \frac{8\pi G m_p}{3cH_0}\,\Omega_b \int_0^z \frac{(1+z')\,f_{\mathrm{IGM}}}{E(z')}\,dz'. \tag{2}$$
4. Calculate and plot the DM as a function of redshift (z) for Planck Cosmology (the numbers for Planck cosmological parameters are found in Planck 2018 paper and associated tables available online) and interpret the results.
5. Vary the values for H_o, Ω_b and Ω_m to some $\pm 10\%$ and interpret how the curve varies.
6. For an observed DM, include the host galaxy contribution (vary its values like 50 or 100 $pc\ cm^{-3}$) using the relation,
$$\mathrm{DM}_{\mathrm{obs}} = \mathrm{DM}_{\mathrm{IGM}} + \frac{\mathrm{DM}_{\mathrm{host}}}{1+z} \tag{3}$$
7. Create a mock FRB catalog dataset (DM and redshifts drawn from a normal distribution and add noise(mimicking cosmic web fluctuations).

III Data Analysis

1. Download CHIME or other FRB catalogs from the publicly available archives. Learn the data structure, glossary and more about the observations.

2. Make exploratory plots of the data present in the catalog and comment on them (histogram of DM and locations in the sky are important ones to plot) to get an idea of FRBs.

3. Use a number of such datasets and cross-match to find the localized FRBs (with host galaxies resolved) to get redshift information (really hard). Alternatively, use the readily available compiled catalog obtained from the literature (much easier). Answer the question *"Is the late-time baryon density consistent with early-Universe CMB measurements?"*

4. Fit the theoretical model for $DM - z$ scatter plot - by fixing H_0, Ω_m, Ω_Λ and f_{IGM} to Planck values and fit for Ω_b. Compare the result with Planck values. Also fix Ω_b, and fit for f_{IGM}.

5. Explore the sensitivity of the calculation with respect to H_0.

6. Compute residual from the best-fit models and measure variance vs redshift. Discuss the relation to large scale structure growth and connect it to inflation.

IV Deliverables

1. A neatly written report of the work done along with the literature review with proper BibTeX citations.

2. A presentation.

3. The whole code and data used.